

1. Sources

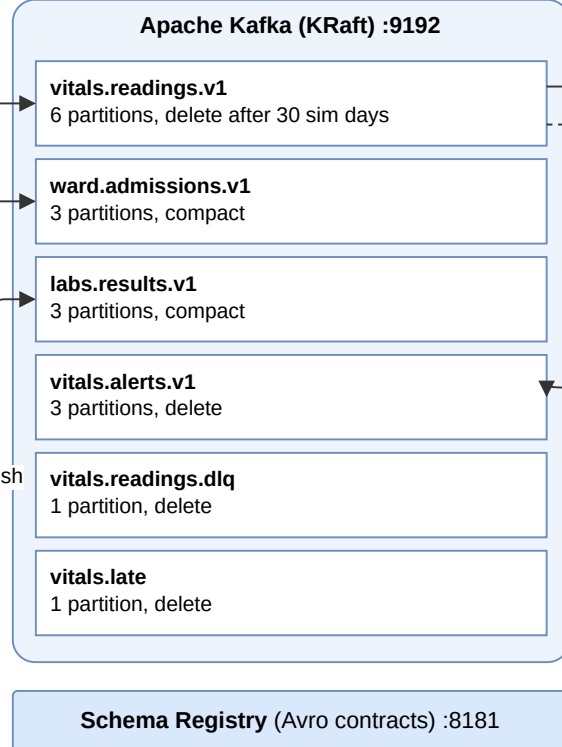
Bedside monitors
40 beds, a reading per bed
every 15 simulated minutes
Python, metrics :8101

Admissions producer
bed, age, condition,
COPD flag

Lab uploader
one JSON file per simulated
day at 06:00 (metrics :8102)

Airflow: ward_lab_ingest
checksum, validate,
quarantine or publish

2. Log (the system of record)



3. Processing (one code path)

ward-stream (NEWS2 v1, live)
Spark Structured Streaming
micro-batch every 5 s
check → 4 h window → NEWS2 → labs → alerts
keeps admissions + labs in a cache
Spark UI :4140, metrics :8104

ward-stream-v2 (NEWS2 v2, replay)
the SAME code, started by
make replay from offset 0
Spark UI :4141, metrics :8105

Apache Airflow :8182 (orchestration only, no scoring)
ward_lab_ingest · ward_daily_report · ward_replay · ward_healthcheck · ward_retention

Observability: Prometheus :9190 scrapes every service's /metrics · Alertmanager :9193 routes safety alerts to the ward · Pushgateway :9191 · kafka-exporter :9309

4. Storage

Apache Cassandra 4.1 :9142

- vitals_by_patient
- risk_scores_by_patient
- ward_risk_snapshot (TTL 120 s)
- alerts_by_ward (TTL 3 h)
- labs_by_patient
- daily_patient_summary
- sim_state

every score row carries its scorer_version

5. Serving

FastAPI :8100
ward list (worst first),
patient, alerts, labs,
replay diff, cutover

Daily PDF report
Airflow ward_daily_report
reads daily_patient_summary

Grafana :3100
ward monitor, pipeline health,
replay v1 vs v2

Colour key

